OPEN SOURCE AGI GRANT / $50,000 / 16 WEEKS

EdgeOML

Measuring whether model fingerprints survive conversion, quantization, and deployment on hardware people actually own.

31/32 BF16 + Q8_0 exact match	**0/256** Decoy matches in each condition	**0.99 GB** Q4_K_M artifact

Thesis. Open models become truly accessible when they run locally. Their identity evidence must survive the same deployment pipeline.

Technical demo complete on Apple M4 / 16 GiB
Proposed public repository: github.com/hex-aragon/edgeoml
Prepared 03 September 2026

1. The missing deployment evidence

Sentient's public OML implementation establishes a key-response model fingerprinting workflow. Real on-device deployment then changes format, runtime, prompt handling, and numeric precision. Today, a failed fingerprint check cannot cleanly identify which transformation caused the failure.

Core question
Does an OML-fingerprinted open model remain identifiable after HF-to-GGUF conversion and practical 8/6/5/4/3/2-bit quantization on laptops and phones?

The controlled comparison

HF BF16	->	**GGUF BF16**	Isolates conversion, runtime, and template effects
GGUF BF16	->	**GGUF Q4**	Isolates quantization effects

Why this fits Sentient

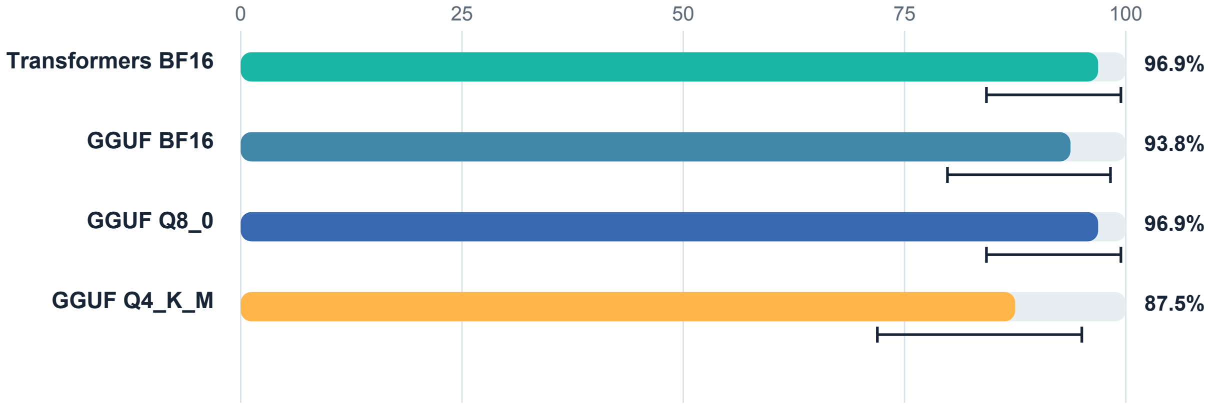
- Personal on-device AI: evaluates models on inexpensive consumer hardware.
- Proof that a model is what it claims: measures lineage separately from artifact identity.
- Agent identity: binds model, runtime, policy, tools, request, and response in an evidence receipt.
- Privacy by default: supports local inference while documenting what is and is not attested.

Claims we deliberately avoid

- A signed model hash is not proof that those weights performed the inference.
- A fingerprint is not unremovable; we will publish removal cost versus utility damage.
- OML does not make payment unbreakable on a fully user-controlled offline device.

2. A real pre-grant pilot

We completed the minimum technical gate on an Apple M4 with 16 GiB unified memory. A pinned Apache-2.0 Qwen2.5 1.5B model received 32 synthetic one-token key-response fingerprints through rank-16 LoRA, was fused, converted to GGUF BF16, and quantized to Q8_0 and Q4_K_M.



Bars: exact-match rate. Whiskers: Wilson 95% confidence interval.

Condition	Size	Exact	Rate	Wilson 95% CI
Transformers BF16	3.09 GB	31/32	96.88%	84.26%-99.45%
GGUF BF16	3.09 GB	30/32	93.75%	79.85%-98.27%
GGUF Q8_0	1.65 GB	31/32	96.88%	84.26%-99.45%
GGUF Q4_K_M	0.99 GB	28/32	87.50%	71.93%-95.03%

0 / 256

Assigned decoy matches in every deployment condition

Wilson 95% upper bound: 1.48%

0 / 32

Original-model accidental matches

Wilson 95% upper bound: 10.72%

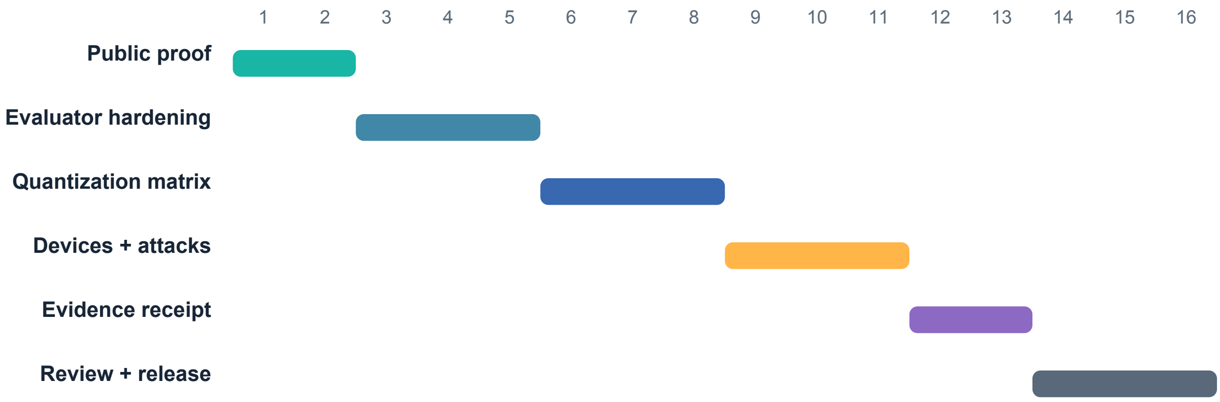
Reproducibility evidence

- 1,184 raw JSONL prediction records and generated Wilson intervals.
- Pinned model, llama.cpp, OML audit revision, package versions, seed, and commands.
- SHA-256 verification for base, adapter, fused, BF16, Q8_0, and Q4_K_M artifacts.
- The failed 640-update checkpoint (7/32) is retained rather than hidden.

Boundary: this is an OML-style MLX LoRA feasibility run, not a reproduction of Sentient's full OML fine-tuning method. The funded work closes that gap.

3. What $50,000 unlocks

The pilot answers whether the idea is technically credible. The grant funds the replication, device diversity, removal attacks, and independent scrutiny required to turn one result into public infrastructure.



16-week acceptance targets

- Two Apache-2.0 model families x three seeds x full practical quantization matrix.
- 144 primary inference cells, each with a manifest, hash, raw records, and result or documented failure.
- Apple reference hardware plus mid-range and flagship Android device measurements.
- Repeated quantization and fine-tuning removal experiments with utility-damage curves.
- L0 receipt schema and Android L1 device-attested prototype, with independent review.

Budget

Category	Amount	Purpose
Research and engineering	$25,000	Harness, devices, report
Reproducible compute	$7,000	Six checkpoints + reruns
Physical test devices	$5,000	Two Android classes
Independent review	$5,000	Security + statistical methods
Documentation + release	$3,000	Guides, demo, community
Hosting + CI	$2,000	Artifacts and runners
Contingency	$3,000	Failed runs and format changes
Total	**$50,000**	**16-week public build**

If Sentient provides compute credits, cash compute savings will be redirected to more negative controls, a third low-cost Android device, and additional independent review.

4. Trust framework and path to adoption

EdgeOML keeps model lineage, artifact identity, agent identity, and execution integrity separate. One signature cannot honestly answer all four questions.

Level	Identity	Adds	Honest boundary
L0	Self-declared	Signed artifact and policy hashes	No proof the hashed model ran
L1	Device-attested	Hardware-backed receipt key	Still trusts user-space measurement
L2	Measured load	Trusted model/runtime measurement	Requires platform or OEM integration
L3	Protected execution	Attested inference or proof	Out of scope for first grant

Commercial validation after the grant

A privacy-first shopping agent will keep preferences, budgets, purchase history, and risk rules on-device. Recommendation receipts can disclose the model and quantization, policy version, offers considered, affiliate commission, selected item, and whether user constraints were satisfied. Commission neutrality becomes a measurable product trust metric, not a marketing promise.

Capital path
$50k open-source grant -> OML deployment evidence -> OEM/TEE measured-load research -> on-device shopping pilot -> investment track with usage, savings, conversion, and bias metrics.

Immediate submission status

- Technical pilot and local evidence package: complete.
- Public repository and stable demo URL: prepared for publication.
- Application request: Open Source AGI Grant Track, $50,000, 16 weeks.

Primary sources

sentient.foundation/grants - program scope
github.com/sentient-agi/OML-1.0-Fingerprinting - public OML implementation
huggingface.co/Qwen/Qwen2.5-1.5B-Instruct - Apache-2.0 base model
github.com/ml-explore/mlx-lm - Apple Silicon LoRA runtime
github.com/ggml-org/llama.cpp - GGUF conversion and quantization

EdgeOML is designed to publish useful evidence whether fingerprints survive Q4 or fail at a reproducible boundary.